

Comprovante do artigo JAS 2018 - página/capa bibliográfica

Categoria: Artigos

Regra: 2.9 - 15 pontos por publicação

Pontos lançados: 15.00

Taylor & Francis Online Journals Search Publish

Home All Journals Mathematics, Statistics & Data Science Journal of Applied Statistics List of Issues Volume 46, Issue 3 Defective models induced by gamma frailty

Journal of Applied Statistics > Volume 46, 2019 - Issue 3

Submit an article Journal homepage

Enter keywords, authors, DOI, etc This Journal Advanced search

327 Views
25 CrossRef citations to date
0 Altmetric

Articles

Defective models induced by gamma frailty term for survival data with cured fraction

Juliana Scudilio , Vinicius F. Calsavara , Ricardo Rocha , Francisco Louzada , Vera Tomazella  & Agatha S. Rodrigues 

Pages 484-507 | Received 03 Aug 2017, Accepted 28 Jun 2018, Published online: 18 Jul 2018

Cite this article  <https://doi.org/10.1080/02664763.2018.1498464> 

Full Article Figures & data References Citations Metrics Reprints & Permissions Read this article Share

Sample our Mathematics & Statistics Journals

>> [Sign in here](#) to start your access to the latest two volumes for 14 days

ABSTRACT

In this paper, we propose a defective model induced by a frailty term for modeling the proportion of cured. Unlike most of the cure rate models, defective models have advantage of modeling the cure rate without adding any extra parameter in model. The introduction of an unobserved heterogeneity among individuals has bring advantages for the estimated model. The influence of unobserved covariates is incorporated using a proportional hazard model. The frailty term assumed to follow a gamma distribution is introduced on the hazard rate to control the unobservable heterogeneity of the patients. We assume that the baseline distribution follows a Gompertz and inverse Gaussian defective distributions. Thus we propose and discuss two defective distributions: the defective gamma-Gompertz and gamma-inverse Gaussian regression models. Simulation studies are performed to verify the asymptotic properties of the maximum likelihood

Related research

People also read

Recommended articles

Cited by 25

The Marshall–Olkin generalized defective Gompertz distribution for surviving fraction modeling >

Taslime Hamdeni et al.
Communications in Statistics - Simulation and Computation
Published online: 14 Aug 2020